Module 9 Assignment

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. Given a nonempty subset S of an inner product space V, show that the set of all vectors in V orthogonal to S, that is

   $$\{x \in V : \langle x, y\rangle = 0 \text{ for all } y \in S\}.$$

   is a subspace of V.

2. Prove that if a matrix $A \in \mathbb{R}^{n\times n}$ is orthogonal, then $\langle Ax, Ay\rangle = \langle x, y\rangle$ for all $x, y \in \mathbb{R}^{n\times 1}$ where $\langle \cdot, \cdot \rangle$ is the standard inner product.

3. (variation on NS 4.7.3) Suppose we wish to find a polynomial $p(t)$ of degree at most 3 with real coefficients such that $p(0) = 0 = p'(0)$ and $\int_0^1 |p(t) - 2 - 3t|^2 \, dt$ is as small as possible.

   Explain how to interpret this as a best approximation problem by identifying the inner product space V, subspace U, vector v to be approximated, and inner product (and associated norm) as described in Definition 4.38. Then find the desired polynomial $p(t)$.

   You must set-up all relevant intregals in your solution, but you may use a computational aid (e.g., Wolfram Alpha) to evaluate the integrals if you wish.

4. Consider

   $$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 1 & 1 & -3 \\ 0 & 1 & 1 \end{bmatrix} \qquad \text{and} \qquad b = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

   (a) Find the QR-factorization of A.

   (b) Find a least squares solution to $Ax = b$.



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 4.3: 2, 3, 5
- Section 4.4: 1, 3, 5, 7, 8, 9, 10
- Section 4.5: 1, 2, 4
- Section 4.7: 1, 2, 3, 5, 6
- Section 4.9: 14, 16

1. Let V be an inner product space and let U be a subspace of V. Show that $T : V \to V$ defined by $T(x) = \text{proj}_U\, x$ is a linear operator.
2. Let $S = \{v_1, \ldots, v_n\}$ be an orthogonal basis for an inner product space V. Show that if $x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$ for $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$, then

$$\|x\|^2 = |\alpha_1|^2\|v_1\|^2 + |\alpha_2|^2\|v_2\|^2 + \cdots + |\alpha_n|^2\|v_n\|^2.$$

